Leptospira Medium Base EMJH
Leptospira Enrichment EMJH

Intended Use

Leptospira Medium Base EMJH is used with Leptospira Enrichment EMJH in cultivating *Leptospira*.

Summary and Explanation

In 1816, Adolf Weil described the first recognized leptospiral infections in humans.[1] These cases were caused by *Leptospira icterohaemorrhagiae* and the disease was subsequently named Weil's Disease.[1] Leptospirosis is a zoonotic disease, having its reservoir in wild, domestic and peridomestic animals. Infection usually results from direct or indirect exposure to the urine of leptospiruric animals.[2] Indirect exposure through contaminated water and soil accounts for most sporadic cases. Direct exposure occurs in pet owners, veterinarians and persons working with livestock.[3]

The basal medium and enrichment are prepared according to the formulations described by Ellinghausen and McCullough[4] as modified by Johnson and Harris.[5] They modified the formula by replacing rabbit serum medium with polysorbate

User Quality Control

Identity Specifications

Difco™ Leptospira Medium Base EMJH

Dehydrated Appearance:	White, free-flowing, homogeneous.
Solution:	2.3 g of base in 900 mL purified water; soluble upon agitation. Solution is colorless, clear.
Prepared Appearance (with Enrichment):	Very light to light amber, clear.
Reaction (Basal Medium) at 25°C:	pH 7.5 ± 0.2

Difco™ Leptospira Enrichment EMJH

Appearance:	Medium to dark amber, clear to very slightly opalescent.

Cultural Response

Difco™ Leptospira Medium EMJH with Enrichment

Prepare the medium per label directions. Inoculate tubes with undiluted fresh cultures of *Leptospira* and incubate at 30 ± 2°C for up to 7 days.

ORGANISM	ATCC™	INOCULUM	RECOVERY
Leptospira interrogans serotype australis	23605	2-3 Loopfuls	Good
Leptospira interrogans serotype canicola	23470	2-3 Loopfuls	Good
Leptospira kirschneri serotype grippotyphosa	23604	2-3 Loopfuls	Good



Uninoculated Tube



Leptospira interrogans serotype australis ATCC™ 23605

80-albumin. Leptospira Medium EMJH was used in cultivation studies of *Leptospira*.[6]

Leptospira Medium EMJH is recommended for the clinical isolation of *Leptospira* from clinical specimens[7] and environmental samples.[8]

Principles of the Procedure

Leptospira Medium Base EMJH contains ammonium chloride, a nitrogen source, and thiamine, a growth factor. Disodium phosphate and monopotassium phosphate are buffering agents. Sodium chloride provides essential ions.

Leptospira Enrichment EMJH contains albumin, polysorbate 80 and additional growth factors for *Leptospira*.

Formulae

Difco™ Leptospira Medium Base EMJH

Approximate Formula* Per Liter

Disodium Phosphate	1.0	g
Monopotassium Phosphate	0.3	g
Sodium Chloride	1.0	g
Ammonium Chloride	0.25	g
Thiamine	5.0	mg

Difco™ Leptospira Enrichment EMJH

A solution of albumin, polysorbate 80 and additional growth factors for *Leptospira*.

**Adjusted and/or supplemented as required to meet performance criteria.*

Directions for Preparation from Dehydrated Product

1. Dissolve 2.3 g of the powder in 900 mL of purified water.
2. Autoclave at 121°C for 15 minutes.
3. Aseptically add 100 mL Leptospira Enrichment EMJH to the medium at room temperature. Mix thoroughly.
4. Test samples of the finished product for performance using stable, typical control cultures.

Procedure[7]

Blood and Cerebrospinal Fluid

Freshly drawn blood is preferable; otherwise, use blood taken with SPS, sodium oxalate or heparin.

1. Inoculate four 5 mL tubes of Leptospira Medium EMJH with 1-2 drops of fluid per tube.
2. Incubate in the dark at 28-30°C or at room temperature.

Urine

A total of 12 tubes will be inoculated for each urine specimen.

1. Prepare 1:10 and 1:100 dilutions of urine using Leptospira Medium EMJH to dilute potential inhibitory substances.
2. Inoculate two 5 mL tubes each of Leptospira Medium EMJH with:
   Urine undiluted, 1 drop per tube;
   Urine diluted 1:10, 1 drop per tube;
   Urine diluted 1:100, 1 drop per tube.
3. Duplicate the above inoculations using medium containing 200 µg/mL 5-fluorouracil to inhibit contaminants.
4. Incubate the tubes in the dark at 28-30°C or at room temperature.

Expected Results[7]

1. Examine tubes weekly for signs of growth (turbidity, haze or a ring of growth).
2. Examine tubes microscopically each week. Take a small drop from a few millimeters below the surface, and examine it with dark-field illumination. Use 400 × magnification.
3. Leptospires will be seen as tightly coiled spirochetes about 1 µm wide and 6-20 µm long. Leptospires rotate rapidly on their long axes and usually have hooked ends.
4. If the specimen is positive, subculture about 0.5 mL taken from the area of growth to two tubes of fresh medium.

References

1. Elliott. 1980. J. Am. Med. Tech. *42*:37.
2. Faine (ed.). 1982. Guidelines for the control of leptospirosis. W. H. O. Offset publication no. 67. World Health Organization, Geneva, Switzerland.
3. Weyant, Bragg and Kaufmann. 1999. *In* Murray, Baron, Pfaller, Tenover and Yolken (ed.), Manual of clinical microbiology, 7th ed. American Society for Microbiology, Washington, D.C.
4. Ellinghausen and McCullough. 1965. Am. J. Vet. Research *26*:45.
5. Johnson and Harris. 1967. J. Bacteriol. *94*:27.
6. Rule and Alexander. 1986. J. Clin. Microbiol. *23*:500.
7. Isenberg and Garcia (ed.). 2004 (update, 2007). Clinical microbiology procedures handbook, 2nd ed. American Society for Microbiology, Washington, D.C.
8. Eaton, Rice and Baird (ed.). 2005. Standard methods for the examination of water and wastewater, 21st ed., online. American Public Health Association, Washington, D.C.

Availability

Difco™ Leptospira Medium Base EMJH

SMWW

Cat. No. 279410 Dehydrated – 500 g

Difco™ Leptospira Enrichment EMJH

SMWW

Cat. No. 279510 Bottle – 6 x 100 mL*

**Store at 2-8°C.*